

# GetSnap User Manual

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Connect an instrument and capture screen images / waveform data with a single hotkey or button press, then save, copy, or paste them.

[Instrument] --Cable(LAN/USB/GPIB)--> [GetSnap] --Hotkey/Button--> [Image/Data] --> [Save to Folder / Clipboard / Excel / CSV]

The screen-capture commands are initial defaults based on each vendor's commonly published command set. They may differ by model/firmware. You can adjust them directly from the app - see "8. Adding & Editing Instruments" below.

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## 1. Launching the App

| Method                 | Action                                    |
|------------------------|-------------------------------------------|
| exe file (recommended) | Double-click dist \ GetSnap.exe           |
| From Python            | venv \ Scripts \ activate → python app.py |

GPIB connections require NI-VISA or Keysight IO Libraries Suite (see "9. Troubleshooting").

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## 2. Screen Layout

| Tab      | Contents                                                                                                                          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------|
| Capture  | Capture buttons (save to folder / copy to clipboard / paste into Excel / save CSV data), Legend, DIV Conditions, preview, history |
| Devices  | VISA resource list, connect                                                                                                       |
| Settings | Save location, background mode, hotkeys, Excel paste, instruments, data export (CSV), Legend, DIV Conditions, language            |

 Save to Folder

 Copy to Clipboard

 Paste into Excel

 Save Data (CSV, etc.)

☒ Enable Legend (applies to folder save & clipboard copy)

|           |  |
|-----------|--|
| MODEL     |  |
| Ope. Mode |  |
| BOM       |  |
| Condition |  |
| Note      |  |

☒ Enable measurement conditions (shows enabled channels' voltage range on the right of the image)

Legend to Display in the Image

|     |     |   |                                                                                     |                                                                                     |
|-----|-----|---|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| CH1 | Vds | V |  |  |
| CH2 | Vgs | V |  |  |

+ Add Channel



No Snap

## History

Detected VISA Resources

↻ Refresh

✓ Connect

Manually enter a resource string (if not auto-detected)

e.g. GPIB0::7::INSTR

→ Manual Connect

Connected: TESTSCOPE,MODEL,SN001,FW1.0 [TestScope]

The folder-icon button at the right end of the tab bar opens the save destination folder directly in Explorer at any time (it's created automatically first if it doesn't exist yet).

### 3. Basic Operation

| Step | Action |
|------|--------|
|------|--------|

|           |                                                                                                                              |
|-----------|------------------------------------------------------------------------------------------------------------------------------|
| ① Connect | "Devices" tab → "Refresh" → select target → "Connect"                                                                        |
| ② Capture | Hotkey, or the buttons on the "Capture" tab                                                                                  |
| ③ Check   | Review the preview/history on the "Capture" tab                                                                              |
| ④ Use it  | Save to folder / copy to clipboard / paste into Excel / save CSV data (see "5. Pasting into Excel" and "6. Saving CSV Data") |

## Hotkeys

The "Settings" tab lets you turn each of the three hotkeys on or off individually, and freely reassign the key combination for each (defaults below). If a hotkey conflicts with another app, just turn that one off and use the buttons on the "Capture" tab instead.

| Purpose           | Default key |
|-------------------|-------------|
| Save to folder    | Ctrl+Alt+F1 |
| Copy to clipboard | Ctrl+W      |
| Paste into Excel  | Ctrl+Alt+F2 |

## When an Error Occurs

The [①...] shown at the start of the error dialog indicates which step failed.

| Step shown                                                           | Meaning                                                                           |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Instrument not connected<br>/ Preset not registered                  | Not yet connected on the "Devices" tab, or no preset is registered                |
| ①Connecting to VISA resource / ①Sending command / receiving response | Communication issue (cable, address, VISA runtime)                                |
| ②Identifying model /<br>②Decoding image                              | Response data format issue                                                        |
| ③Post-processing                                                     | Image processing issue (background conversion, Legend/DIV Conditions compositing) |
| ④Saving / ④Copying                                                   | Permissions or disk space issue at the save destination                           |

|                                                                  |                                                                                             |
|------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Data export not supported                                        | "Save CSV Data" was pressed, but the connected model has no data export commands registered |
| ① Sending data-export command to instrument / receiving response | Communication issue during CSV data export (cable, commands)                                |
| ② Saving data file                                               | Data retrieval succeeded; issue with permissions or disk space at the save destination      |

## 4. Adding Information to the Image: Legend & DIV Conditions

Two toggles on the Capture tab let you auto-composite information onto the image. Both apply to folder save, clipboard copy, and pasting into Excel alike.

☒ Enable Legend (applies to folder save & clipboard copy)

| MODEL     |  |
|-----------|--|
| Ope. Mode |  |
| BOM       |  |
| Condition |  |
| Note      |  |

☒ Enable measurement conditions (shows enabled channels' voltage range on the right of the image)

Legend to Display in the Image

|     |     |   |                                                                                                   |                                                                               |
|-----|-----|---|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| CH1 | Vds | V | <span style="background-color: yellow; width: 20px; height: 15px; display: inline-block;"></span> | <span style="background-color: red; color: white; padding: 2px 5px;">✕</span> |
| CH2 | Vgs | V | <span style="background-color: green; width: 20px; height: 15px; display: inline-block;"></span>  | <span style="background-color: red; color: white; padding: 2px 5px;">✕</span> |

+ Add Channel

### 4.1 Legend: free-text fields attached to the image

Turning on "Enable Legend" shows a list of items on the Capture tab (e.g. MODEL, Ope. Mode, BOM, Condition, Note). Fill in values and capture - they're composited as a table at the top of the image. Adding/removing items, reordering, and font settings (font, Bold, Italic) are managed from the "Legend" section of the "Settings" tab (not covered by "8. Adding & Editing Instruments").

### 4.2 DIV Conditions: per-channel voltage range, shown automatically

Turning on "Enable measurement conditions" lets you pick which channels (CH1-8) / Math traces (M1-8) to display, and enter a custom legend name per row (e.g. CH1: Vds). On every capture, GetSnap queries the instrument and composites the actual voltage range (e.g. 5 V/div) into a panel whose row backgrounds match each channel's waveform color.

| Action                        | Steps                                                                                                                                                                                                                                    |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Add a channel                 | " + Add Channel" → choose CH1-8 / M1-8 from the dropdown                                                                                                                                                                                 |
| Edit the legend name          | Type into each row's text field (leave blank to show just the ID, e.g. "CH1"). Press "Enter" or "Shift+Enter" to add line breaks for a multi-line legend name                                                                            |
| Edit the unit                 | Type directly into each row's unit field (default V). For a current probe, for example, enter A - any unit other than voltage can be entered freely (there is no way to query the unit from the instrument, so it's always set manually) |
| Remove a channel              | Red trash-can icon at the right of each row                                                                                                                                                                                              |
| Change the insertion position | "Settings" tab → "DIV Conditions Position" (Right / Left / Bottom / Top)                                                                                                                                                                 |
| Change the font               | "Settings" tab → "DIV Conditions Font" (Bold/Italic can also be set)                                                                                                                                                                     |

Choosing "Top" inserts the DIV Conditions panel between the Legend and the waveform image (stacking order: Legend → DIV Conditions → waveform).

Example output (CH1=Vds, CH2=Vgs shown on the right):



### 4.3 Choosing channel colors (always manual)

Click the color button (the square swatch to the right of each legend-name field) to pick a color from the palette manually (hex input also supported). Any channel you haven't given a color falls back to the default palette (CH1=yellow, CH2=green, etc., matching common oscilloscope conventions).

Automatically reading display colors from the instrument was removed: it required more commands when adding a new supported instrument and didn't generalize well across vendors. Colors are always set one of the two ways above.

#### 4.4 If the voltage range query fails

If reading a channel's DIV condition voltage range fails, the capture itself is never aborted. An error message is shown, and the image is saved/copied without the DIV Conditions panel (the same appearance as when the feature is disabled). If you keep seeing this error, check the command setup in the "Support DIV condition display" card - see "8.7 DIV Conditions commands".

#### 4.5 Adding the time axis (horizontal) range

Turning on "Enable time axis display" under "Settings" tab → "DIV Conditions" adds a row for the time axis (horizontal) range (e.g. 1 ms/div) at the front of the DIV Conditions panel.

| Action                        | Steps                                                          |
|-------------------------------|----------------------------------------------------------------|
| Turn time axis display on/off | "Settings" tab → "DIV Conditions" → "Enable time axis display" |

Non-oscilloscope instruments (VNAs, spectrum analyzers, etc.) don't have a time-based horizontal axis, so while "Enable time axis display" is OFF, GetSnap never sends a time axis range command to the instrument (to avoid errors on instruments that don't support it). The command is only sent once you turn it ON, and if the query fails, the capture itself is never aborted (same as 4.4 - an error is shown and the image is saved without the time axis information). You can check the command setup in "8.7 DIV Conditions commands".

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### 5. Pasting into Excel

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Pressing "Paste into Excel" (or its hotkey) inserts the image as a picture shape directly at the currently selected cell in Excel. Unlike a clipboard-based paste, the shape size is specified in points, so it's unaffected by the sheet's zoom level, and it's always generated from the full-resolution image right after capture, so there's no quality loss.

| Setting                            | Details                                                                                                                       |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Image width when pasted into Excel | "Settings" tab → "Excel Paste" section (default 80mm; height is computed automatically to preserve the original aspect ratio) |

If Excel isn't running, or no cell is currently selected, nothing happens (when triggered by hotkey this is only a brief status-bar note - no error dialog is shown).

## 6. Saving CSV Data

Pressing the green "Save Data (CSV, etc.)" button on the "Capture" tab runs the connected model's waveform data commands and saves the result as a separate CSV file, alongside the image, in the same save folder.



### 6.1 Automatically fetching every enabled channel

If the connected model's preset has a "channel enabled-check command" configured, GetSnap automatically detects every channel currently displayed (enabled) on the oscilloscope and combines them into a single CSV (SampleIndex, CH1, CH2, ... columns). Models without this command configured fetch CH1 only (see "8.6 CSV data export commands").

### 6.2 Automatic decimation for long record lengths

On models with long record lengths, the CSV file size can reach tens of MB or millions of rows - too large to open in Excel (max ~1.04M rows per sheet). Only when the point count exceeds the "Max Points for CSV Export" setting (default 50,000; "Settings" tab → "Data Export") does GetSnap automatically decimate it, while preserving each channel's and each segment's minimum and maximum values. Whether or not decimation happens, a `SampleIndex` column (the original sample position) is always included, so the time-series information is never lost.

### 6.3 Cancelling a save in progress

While saving, the status bar shows an animated progress indicator and a "X" button. On models with long record lengths, retrieval can take tens of seconds to a few minutes - press "X" if you want to cancel.

VISA communication has no safe cancellation mechanism, so pressing "X" forcibly disconnects the instrument to interrupt the operation. After cancelling, reconnect from the "Devices" tab.

## 7. Supported Instruments

The actual commands can always be reviewed and edited from the dialog in "8. Adding & Editing Instruments" below.

| Vendor          | Image Format | CSV Data Export | Notes                                                                                                                                                                    |
|-----------------|--------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tektronix       | PNG          | Yes             | -                                                                                                                                                                        |
| Yokogawa        | PNG          | Yes             | Return-to-local uses GTL only (verified on real hardware, see 8.5)                                                                                                       |
| LeCroy          | PNG          | No              | Not verified on real hardware (initial implementation based on the official manual only)                                                                                 |
| Keysight E5061B | PNG          | Yes             | Verified on real hardware. DIV Conditions doesn't apply (it's a network analyzer). CSV export fetches only the selected trace per channel (CH1 only by default, see 8.6) |

The built-in presets currently cover these four vendors. Other instruments can be added via "8.3 Adding a new instrument".

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## 8. Adding & Editing Instruments

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Go to "Settings" → "Instrument Settings..." to add or edit supported instruments. Your edits persist across app restarts.

Save

Save Folder

C:\Users\daisu\GetSnap Captures

Browse...

Background Conversion Mode

Invert (B/W)

Hotkeys

Same actions are also available from the buttons in the "Capture" tab

☒ Save to Folder

Ctrl+Alt+F1

☒ Copy to Clipboard

Ctrl+W

☒ Paste into Excel

Ctrl+Alt+F2

Excel Paste

Image Width When Pasted into Excel

80.00 mm

Add/Edit Instruments

See Manual\_ENG.pdf for the list of supported vendors and commands.

Instrument Settings...

Data Export

On models with long record lengths, the file size may become too large to open in Excel.

Max Points for CSV Export

50000

Legend

Legend items (edit label / remove)

MODEL

Ope. Mode

BOM

Condition

Note

++ + Add Item

Legend Font

メイリオ (Meiryo)

15

Bold

Italic

DIV Conditions

DIV Conditions Position

Right

DIV Conditions Font

メイリオ (Meiryo)

15

Bold

Italic

☒ Enable time axis display

Display

Language / 言語

English

## 8.1 First, find the model match string

Before adding a new instrument, connect to it once from the "Devices" tab. You can connect even without a registered preset - the `*IDN?` response string is shown as-is on screen (e.g. Connected: ANRITSU,MS2840A,SN001,FW2.0 [Unsupported model (no preset registered)]). Use part of that string (e.g. the vendor name) as the "model match string".

## 8.2 Editing an existing preset

Select an instrument from the list to open its form on the right. Edit the fields and press "Save" to override that instrument's settings only. Built-in presets cannot be renamed. There is no button to revert an edited preset back to its original content, so back up a built-in preset with "Export" before editing it (see 8.8).

Registered instruments

Tektronix

Yokogawa

LeCroy

Keysight E5061B

+ Add New

□ Import

☰ Export

Save

Delete

This is a built-in preset. Its name cannot be changed (use "+ Add New" to register it under a different name instead). Editing and saving overrides this instrument's settings only.

[Required] Name

Yokogawa

[Required] Model match string (substring of the "IDN?" response, matched case-insensitively)

yokogawa

[Required] Image save commands (one per line, last line is the query)

:IMAGe:TONE REVERSE  
:IMAGe:FORMat PNG  
:IMAGe:SEND?

Run Test

[Required] Commands (SCPI, one per line)

Sent in addition to the VISA-level GTL command when returning the instrument to local control after a capture or data export. GPIB/USB connections already use GTL as the primary mechanism, so this can usually be left blank.

Optional

Support no-invert capture

Used when "No Invert" is selected as the background mode in the "Settings" tab. Leave this off to reuse the image save commands instead.

[Optional] Commands (one per line, last line is the query)

:IMAGe:TONE COLOr  
:IMAGe:FORMat PNG  
:IMAGe:SEND?

Image Format

Delay Between Commands

PNG

0.00 s

Support CSV data export

Used by the "Save CSV Data" button in the "Capture" tab. If you set a channel enabled-check command, every channel currently displayed (enabled) on the oscilloscope is automatically fetched and combined into a single CSV.

[Optional] Data commands (one per line, last line is the query; "[ch]" is replaced with the channel number)

:STOP  
:WAVeform:TRACe (ch)

Run Test

[Optional] Data export format (e.g. CSV)

CSV

[Optional] Channel enabled-check command ("[ch]" is replaced with the channel number; expects an ON/OFF or 1/0 style response)

## 8.3 Adding a new instrument

Press "+ Add New" at the bottom left to open a blank form. Fill in the required fields (name, model match string, image save commands, return-to-local commands) and press "Save" to add it to the list. "Save" and "Delete" are pinned to the top of the form. The red "Delete" button removes it at any time, after a confirmation popup - deletion cannot be undone (for a built-in preset, it is instead removed from the list). Back it up with "Export" beforehand if you might want it again (see 8.8).

Optional fields are grouped into three feature cards - "Support no-invert capture", "Support CSV data export", and "Support DIV condition display" - and each card's fields only appear once you turn on its checkbox for a feature you actually want to use. Leave a card off and its fields stay hidden, keeping the form focused on what you need.

Registered instruments

Tektronix

Yokogawa

LeCroy

Keysight E5061B

+ Add New

□ Import

☰ Export

SaveDelete

[Required] Name

[Required] Model match string (substring of the "IDN?" response, matched case-insensitively)

e.g. yokogawa

[Required] Image save commands (one per line, last line is the query)

e.g.

:IMAGe:FORMat PNG

:IMAGe:SEND?

Run Test

[Required] Commands (SCPI, one per line)

Sent in addition to the VISA-level GTL command when returning the instrument to local control after a capture or data export. GPIB/USB connections already use GTL as the primary mechanism, so this can usually be left blank.

Optional

☐ Support no-invert capture

Used when "No Invert" is selected as the background mode in the "Settings" tab. Leave this off to reuse the image save commands instead.

Image FormatDelay Between Commands

PNG

0.00 s

☐ Support CSV data export

Used by the "Save CSV Data" button in the "Capture" tab. If you set a channel enabled-check command, every channel currently displayed (enabled) on the oscilloscope is automatically fetched and combined into a single CSV.

☐ Support DIV condition display

Used by the "DIV Conditions" feature to auto-reflect the user-selected channels/Math traces' voltage range (V/div) in the image, and the time axis (horizontal) range when "Enable time axis display" is ON in the Settings tab.

[Optional] Notes

## 8.4 Field Reference (required)

| Field                    | Description                                                                                                   | Example            |
|--------------------------|---------------------------------------------------------------------------------------------------------------|--------------------|
| Name                     | Display name in the list (built-in presets cannot be renamed)                                                 | Anritsu            |
| Model match string       | Substring of the *IDN? response, matched case-insensitively; the first match from the top of the list is used | anritsu            |
| Image save commands      | Screen capture commands. One per line; the last line is the query that returns the image                      | :HCOPy:SDUMp:DATA? |
| Return-to-local commands | Extra command(s) to restore the front panel after capture / data export. Usually left blank (see 8.5)         | Usually blank      |

Image format and Delay between commands are always-valued inputs (combo box / spin box), so they carry no required/optional tag, but both are still used for image capture.

## 8.5 Leave "Return-to-local commands" blank in most cases

For GPIB/USB connections, GetSnap already restores the front panel automatically after capture via VISA-level GTL (Go To Local). Entering a command the instrument doesn't recognize here can actually cause a command-error message or leave the panel locked (confirmed through real hardware testing on a Yokogawa DLM5058). Only enter a command here - one you've verified against the model's programming manual - if GTL alone does not restore the panel for your instrument.

## 8.6 CSV data export commands (optional)

Turning on "Support CSV data export" reveals the following fields.

Registered Instruments

Tektronix

**Yokogawa**

LeCroy

Keysight E5061B

+ Add New

□ Import

📄 Export

SaveDelete

:IMAGE:TONE REverse

:IMAGE:FORMat PNG

:IMAGE:SEND?

Run Test

[Required] Commands (SCPI, one per line)

Sent in addition to the VISA-level GTL command when returning the instrument to local control after a capture or data export. GPIB/USB connections already use GTL as the primary mechanism, so this can usually be left blank.

Optional

✓ Support no-invert capture

Used when "No Invert" is selected as the background mode in the "Settings" tab. Leave this off to reuse the image save commands instead.

[Optional] Commands (one per line, last line is the query)

:IMAGE:TONE COlor

:IMAGE:FORMat PNG

:IMAGE:SEND?

Image Format

Delay Between Commands

PNG

0.00 s

✓ Support CSV data export

Used by the "Save CSV Data" button in the "Capture" tab. If you set a channel enabled-check command, every channel currently displayed (enabled) on the oscilloscope is automatically fetched and combined into a single CSV.

[Optional] Data commands (one per line, last line is the query; "{ch}" is replaced with the channel number)

:STOP

:WAVEform:TRACe {ch}

Run Test

[Optional] Data export format (e.g. CSV)

CSV

[Optional] Channel enabled-check command ("{ch}" is replaced with the channel number; expects an ON/OFF or 1/0 style response)

:CHANnel{ch}:DISPlay?

Max channels to check (CH1 through this number)

4

Support DIV condition display

Used by the "DIV Conditions" feature to auto-reflect the user-selected channels/Math traces' voltage range (V/div) in the image, and the time axis (horizontal) range when "Enable time axis display" is ON in the Settings tab.

[Optional] Channel (CH) voltage range query ({ch} is replaced with the channel number)

| Field                         | Description                                                                                                                     | Example                                |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| Data commands                 | Waveform data retrieval commands. One per line, last line is the query. {ch} is replaced with the channel number (1, 2, 3, ...) | :WAVEform:TRACe {ch} → :WAVEform:SEND? |
| Data export format            | Roughly, the saved file's extension                                                                                             | CSV                                    |
| Channel enabled-check command | {ch} is replaced with the channel number; expects an ON/OFF or 1/0 style response                                               | :CHANnel{ch}:DISPlay?                  |

|                       |                                           |   |
|-----------------------|-------------------------------------------|---|
| Max channels to check | Check CH1 through this number (default 4) | 4 |
|-----------------------|-------------------------------------------|---|

If the channel enabled-check command is left blank, there's no way to tell which channels are enabled, so GetSnap falls back to fetching CH1 only.

### 8.7 DIV Conditions commands (optional)

To use the "4.2 DIV Conditions" feature, turn on "Support DIV condition display" and configure the target instrument's voltage-range query commands. A blank field simply means that part of the feature won't work - it has no effect on image capture itself.

Registered instruments

Tektronix

Yokogawa

LeCroy

Keysight E5061B

+ Add New

Import

Export

SaveDelete

Sent in addition to the VISA-level GTL command when returning the instrument to local control after a capture or data export. GPIB/USB connections already use GTL as the primary mechanism, so this can usually be left blank.

Optional

Support no-Invert capture

Used when "No Invert" is selected as the background mode in the "Settings" tab. Leave this off to reuse the image save commands instead.

[Optional] Commands (one per line, last line is the query)

:IMAGE:TONE COlor  
:IMAGE:FORMat PNG  
:IMAGE:SEND?

Image FormatDelay Between Commands

PNG0.00 s

Support CSV data export

Used by the "Save CSV Data" button in the "Capture" tab. If you set a channel enabled-check command, every channel currently displayed (enabled) on the oscilloscope is automatically fetched and combined into a single CSV.

[Optional] Data commands (one per line, last line is the query; "(ch)" is replaced with the channel number)

:STOP  
:WAVEform:TRACe (ch)

Run Test

[Optional] Data export format (e.g. CSV)

CSV

[Optional] Channel enabled-check command ("(ch)" is replaced with the channel number; expects an ON/OFF or 1/0 style response)

:CHANnel(ch):DISPlay?

Max channels to check (CH1 through this number)

4

Support DIV condition display

Used by the "DIV Conditions" feature to auto-reflect the user-selected channels/Math traces' voltage range (V/div) in the image, and the time axis (horizontal) range when "Enable time axis display" is ON in the Settings tab.

[Optional] Channel (CH) voltage range query ((ch) is replaced with the channel number)

:CHANnel(ch):VDIV?

[Optional] Math trace voltage range query ((ch) is replaced with the trace number)

:MATH(ch):SCALE?

[Optional] Time axis (horizontal) range query ((ch) is not used; one value for the whole instrument)

| Field                            | Description                              | Example            |
|----------------------------------|------------------------------------------|--------------------|
| Channel (CH) voltage range query | {ch} is replaced with the channel number | :CHANnel{ch}:VDIV? |
| Math trace voltage range query   | {ch} is replaced with the trace number   | :MATH{ch}:SCALE?   |

|                                    |                                                                                                                                         |                 |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Time axis (horizontal) range query | {ch} is not used (one value for the whole instrument). Only sent while "Enable time axis display" is ON in the "Settings" tab (see 4.5) | :TImebase:TDIV? |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-----------------|

On some instruments, the Math trace voltage range query returns a different response format than a regular channel - for example a semicolon-separated compound response like :MATH1:SCAL:CENT 0.0000000E+00;MODE AUTO;SENS 2.0000000E+00. In that case GetSnap automatically extracts the numeric value of the SENS field (the per-division sensitivity), so you can still configure it with the same {ch} substitution as a regular channel.

## 8.8 Exporting/importing presets

The "Export" and "Import" buttons (icon-only) at the top of the list let you save a single preset's content to a JSON file, or load one back from a previously saved file.

| Purpose                            | Steps                                                                                                          |
|------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Back up a preset                   | Select it from the list, press "Export", choose where to save                                                  |
| Restore it / move it to another PC | Press "Import", pick the JSON file (you'll be asked to confirm before overwriting a preset with the same name) |

Built-in presets have no "revert to original" button, so always back up a built-in preset with "Export" before editing or deleting it (see 8.2, 8.3).

## 8.9 Testing commands against the instrument

Below both the "Image save commands" field and the "Data commands" field is a "Run Test" button that sends the commands currently typed in that field to the connected instrument and shows you the response. It's disabled while no instrument is connected.

This lets you verify a command works before saving the form, instead of saving first and only finding out it's wrong once you try a real capture. A text response is shown as-is; a binary response (such as image data) is shown as a hex preview of its first bytes.

# 9. Troubleshooting

| Symptom                                         | Fix                                            |
|-------------------------------------------------|------------------------------------------------|
| Instrument doesn't appear in the list over GPIB | Install NI-VISA or Keysight IO Libraries Suite |

|                                                              |                                                                                                                                                                                                                                   |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Connects fine but capture fails                              | Check the failing step at the start of the error, then review commands in "8."                                                                                                                                                    |
| Want to change the image background color                    | Toggle "Background Conversion Mode" on the "Settings" tab                                                                                                                                                                         |
| A hotkey doesn't respond                                     | May conflict with another app; turn just that hotkey off in "Settings" and use the buttons on the "Capture" tab instead                                                                                                           |
| Front panel unresponsive after capture                       | See "8.5 Leave 'Return-to-local commands' blank in most cases"                                                                                                                                                                    |
| DIV Conditions shows an error / doesn't appear               | Check the command setup in "8.7 DIV Conditions commands". On error, the image is still saved without DIV Conditions                                                                                                               |
| Time axis (horizontal) range shows an error / doesn't appear | Check that "Enable time axis display" is ON in the "Settings" tab, and that the time axis range query in "8.7 DIV Conditions commands" is correct (see 4.5). On error, the image is still saved without the time axis information |
| Channel colors don't match the real instrument               | See "4.3 Choosing channel colors" - pick the color manually from each row's color button                                                                                                                                          |
| CSV data export only fetches some channels                   | Check the channel enabled-check command and max channel count in "8.6 CSV data export commands"                                                                                                                                   |
| Cancelling a CSV save left the instrument disconnected       | This is expected (see 6.3) - reconnect from the "Devices" tab                                                                                                                                                                     |